# Week 1: Data Exploration Report

*DePaul University Graduate Applicant Dataset*

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## 1. Dataset Familiarization Summary

### 1.1 Dataset Overview

The DePaul University Graduate Applicant Dataset contains records of individuals who applied to or were contacted about graduate programs at DePaul University. The cleaned dataset provided for analysis comprises:

| Attribute | Detail |
|---|---|
| **Total Records (Rows)** | 5,000 |
| **Total Fields (Columns)** | 45 |
| **Numerical Columns** | 8 (Reference_ID, Received_At, Created_At, Modified_At, and float variants) |
| **Categorical / Text Columns** | 37 (names, country, programme, semester, source, etc.) |
| **Date-related Columns** | 6 (Admit_Date, Date_of_Contact, Start_Date, Created_At variants) |
| **Duplicate Rows** | 0 (no exact duplicate records found) |
| **Unique Applicants (Reference_ID)** | 5,000 (all unique) |

### 1.2 Key Initial Observations

- The dataset spans graduate applicants across multiple years (2022–2026), with the highest volume concentrated in 2024 and 2025.
- A significant block of 63.9% of records (3,194 rows) appears to lack outreach-related information, suggesting these applicants were either not yet contacted or their outreach data resides in a separate system.
- India is overwhelmingly the largest source country, contributing over half of all applicants (53.7%).
- Program demand is heavily skewed toward technology and business disciplines: Business Analytics, Computer Science, and Data Science form the top three programs.
- The dataset is structured across what appear to be two merged tables, an applications table and an outreach/contact table joined on Reference_ID. The high rate of nulls in outreach columns reflects this merge gap.
- All available contact records (Degree_Type-2) are classified as "grad," confirming this is a graduate-level applicant dataset.

## 2. Data Dictionary

The table below documents all 45 dataset fields, their data types, variable definitions, and any missingness noted during exploration.

| Field Name | Data Type | Definition | Missing Values? |
|---|---|---|---|
| **Reference_ID** | Integer | Unique applicant identifier | No |
| **Given_Name** | String | Applicant first name | Yes – 2 missing |
| **Last_Name** | String | Applicant last name | No |
| **College** | String | Applicant undergraduate institution & degree | Yes – 65 missing |
| **Country** | String | Applicant country of origin | Yes – 10 missing |
| **Received_At** | Integer (timestamp) | Application receipt timestamp (epoch ms) | No |
| **University** | String | DePaul University (constant) | No |
| **Intake** | String | Program + intake term combined label | Yes – 3 missing |
| **College_1st_Choice** | String | Preferred DePaul college of application | Yes – 6 missing |
| **Phone_Number** | String | Applicant contact number | No |
| **Street_1** | String | Primary address line | Yes – 13 missing |
| **Street_2** | String | Secondary address line | Yes – 3,246 missing (64.9%) |
| **City** | String | Applicant city of residence | Yes – 30 missing |
| **Region** | String | State or region | No |
| **Postal** | String | Postal/ZIP code | Yes – 190 missing |
| **Email_ID** | String | Applicant email address | Yes – 1 missing |
| **Comments** | String | Admission officer notes | No |
| **Admit_Date** | String | Date of admissions decision | No |
| **Application_Source** | String | Study Group or Non Study Group recruitment channel | No |

| Field Name | Data Type | Definition | Missing Values? |
|---|---|---|---|
| **Created_At** | Integer (timestamp) | Application creation timestamp | No |
| **Modified_At** | Integer (timestamp) | Last modification timestamp | No |
| **Reference_ID-2** | Float | Secondary reference (outreach record) | Yes – 3,194 missing (63.9%) |
| **Date_of_Contact** | String | Date of outreach contact attempt | Yes – 3,194 missing |
| **Caller_Name** | String | Name of outreach caller | Yes – 3,194 missing |
| **Outcome_1** | String | Result/status of outreach call | Yes – 3,194 missing |
| **Remark** | String | First outreach note | Yes – 4,865 missing (97.3%) |
| **Remark_2** | String | Second outreach note | Yes – 4,993 missing (99.9%) |
| **Campaign_Id** | Float | Associated marketing campaign ID | Yes – 3,194 missing |
| **Slate_Form_Filled** | String | Whether Slate admission form was completed | Yes – 3,194 missing |
| **ID** | Float | Secondary record identifier | Yes – 3,194 missing |
| **Name** | String | Alternate applicant name field | Yes – 3,261 missing (65.2%) |
| **Attempts** | Float | Number of contact attempts | Yes – 3,194 missing |
| **Degree_Type-2** | String | Degree level (all "grad" in available records) | Yes – 3,194 missing |
| **Type** | String | Application type classification | Yes – 3,194 missing |
| **Start_Date** | String | Programme start date | Yes – 3,194 missing |
| **Program_Name** | String | Name of academic programme applied for | Yes – 3 missing |
| **Semester** | String | Intake semester (Fall/Winter/Spring/Summer) | Yes – 3 missing |
| **Year** | Float | Intake year | Yes – 3 missing |

**Note on Ambiguous Variables:** The dataset contains multiple duplicated column names with suffixes (-2, -3) such as Reference_ID-2, Created_At-2, and University-2/3. These appear to originate from separate source tables merged into a single flat file. Fields like Remark and Remark_2 have near-total missingness (97–99%) and may be legacy annotation fields. The Received_At and Created_At fields are stored as epoch milliseconds and require conversion for meaningful time analysis.

# 3. Data Quality Report

## 3.1 Missing Values Summary

The dataset contains several fields with missing values, ranging from minor gaps to structural absences. The table below summaries the most significant:

| Field | Missing Count | Missing % | Interpretation |
|---|---|---|---|
| **Remark_2** | 4,993 | 99.9% | Effectively empty – likely a legacy field |
| **Remark** | 4,865 | 97.3% | Sparse annotation field; not reliable for analysis |
| **Name** | 3,261 | 65.2% | Alternate name column; overlaps with Given_Name/Last_Name |
| **Street_2** | 3,246 | 64.9% | Normal for address field (not all applicants have Apt/Suite) |
| **Outreach Block (16 fields)** | 3,194 | 63.9% | Applicants not linked to outreach records structural gap |
| **Postal** | 190 | 3.8% | Minor; may reflect international applicants with non-standard codes |
| **College** | 65 | 1.3% | Low manageable with imputation or exclusion |
| **Country** | 10 | 0.2% | Negligible; likely data entry omissions |
| **Email_ID** | 1 | 0.02% | Single missing records |
| **Given_name** | 2 | 0.04% | Single missing records |

## 3.2 Duplicates & Inconsistencies

- No duplicate rows were identified in the dataset.
- All 5,000 Reference_ID values are unique — strong primary key integrity.
- The Intake column combines Program Name and Semester/Year into a single string. This field is split downstream into Program_Name, Semester, and Year columns, which is better practice for analysis.
- Attempts field has only 1,806 populated values, all equal to 1.0, indicating no variation and limited analytical utility in its current form.
- The University column appears to be a constant ("DePaul University") and provides no analytical value; similarly, University-2 and University-3 are redundant.
- Timestamps (Received_At, Created_At, Modified_At) are stored as epoch milliseconds, these need conversion before trend analysis.

## 3.3 Identified Anomalies

- Duplicate-style columns from apparent table merge: Reference_ID-2, Received_At-2, Created_At-2/3, Modified_At-2/3, and University-2/3 suggest a poorly structured flat-file export.
- Year field includes "2022" (1 record) and "2026" (41 records), which may be outliers or data entry errors warranting review.
- No data type conflicts were found, all string fields are cleanly stored as text and numeric fields as integers or floats.

## 4. Exploratory Data Analysis (EDA) & Visualisations

### 4.1 Applicant Country Distribution

India is the single largest source country by a substantial margin, contributing 2,684 applicants (53.7% of the dataset). The United States follows at 9.8%, and African countries Ghana (8.2%) and Nigeria (7.2%) collectively represent a meaningful share of the applicant pool. The top 10 countries together account for approximately 91% of all applicants.

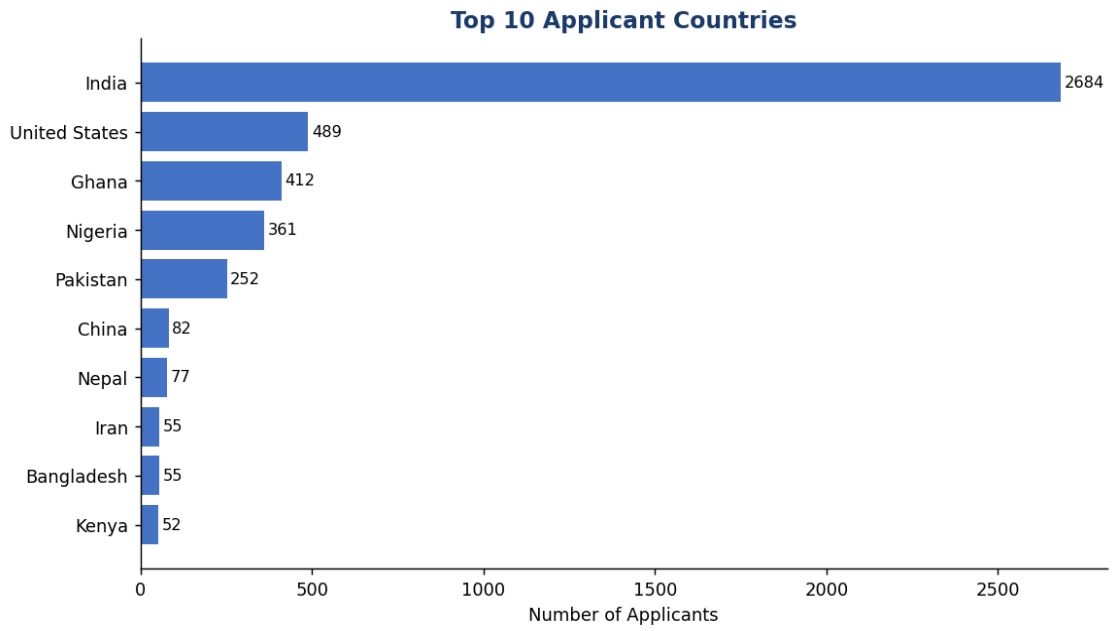



*Figure 1: Top 10 Applicant Countries by Volume*

### 4.2 College First Choice Distribution

Applicants strongly prefer the Jarvis College of Computing and Digital Media (48.3%), followed by the Kellstadt Graduate School of Business (32.8%). These two colleges together represent over 81% of all first-choice preferences, reflecting the tech and business orientation of the applicant pool.

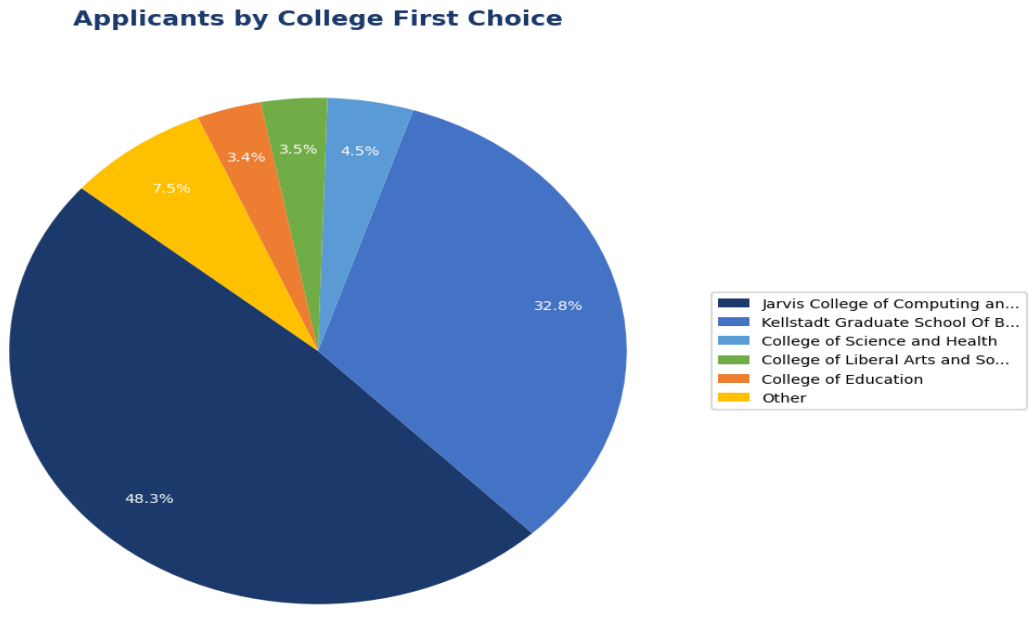



*Figure 2: College First Choice Top 5 and Others (Pie Chart)*

### 4.3 Applications by Intake Semester

Fall is by far the most popular intake period, accounting for 59.7% of all applications (2,983). Winter is a distant second at 25.8% (1,288). Spring and Summer together represent only 14.5% of applications. This seasonality likely reflects standard international student admission cycles aligned with September starts.

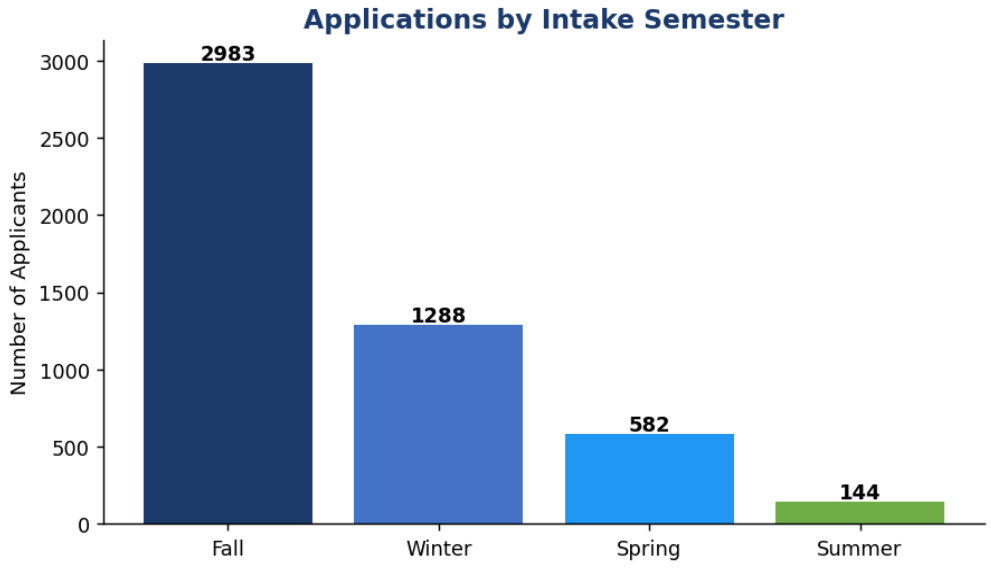



*Figure 3: Application Volume by Intake Semester*

### 4.4 Top 10 Programs by Applicant Count

Business Analytics leads with 707 applicants (14.1%), closely followed by Computer Science at 607 (12.1%) and Data Science at 337 (6.7%). The top five programmes are all STEM or data-adjacent disciplines, consistent with the global demand for tech talent. Health Informatics and Cybersecurity rank 4th and 5th respectively.

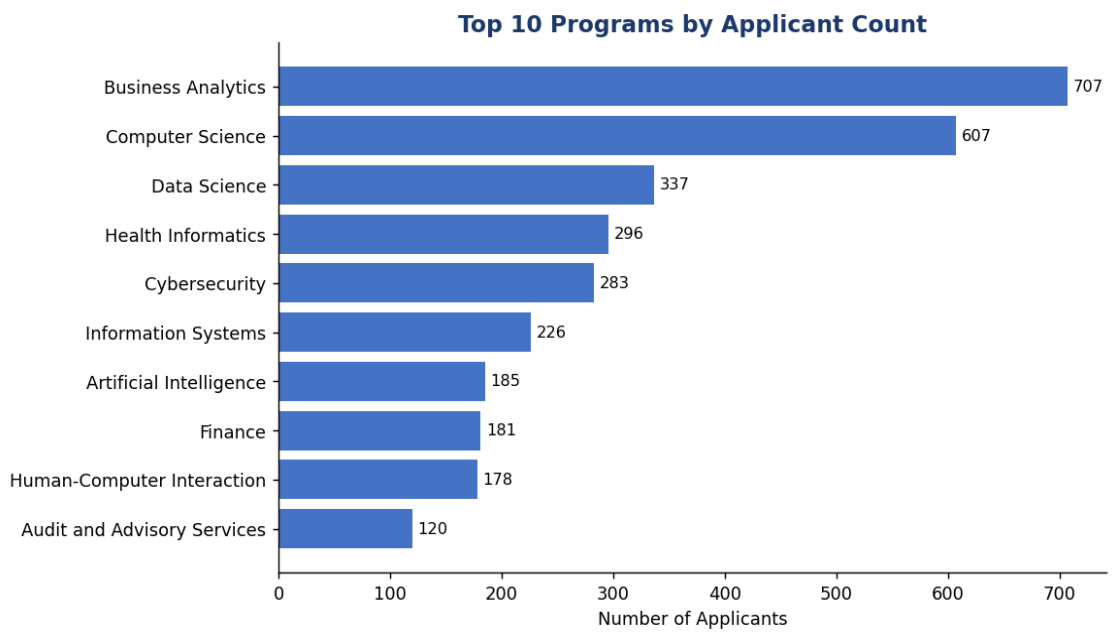



*Figure 4: Top 10 Academic Programs by Applicant Count*

## 4.5 Application Source Distribution

The overwhelming majority of applicants 84.2% (4,212) came through Non Study Group channels, while only 15.8% (788) were Study Group applicants. This suggests that DePaul's direct recruitment and online/organic application pipelines significantly outperform Study Group referrals in volume.

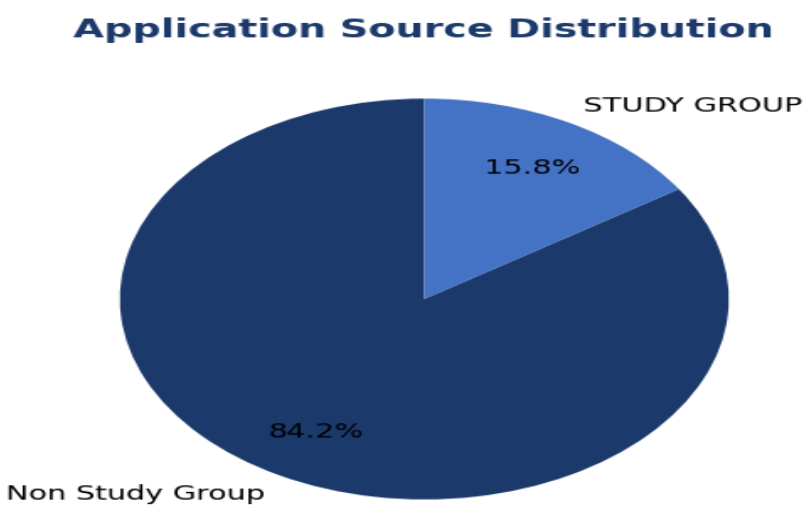



*Figure 5: Application Source  Study Group vs Non Study Group*

## 4.6 Application Volume Trend by Year

Application volume grew substantially from 227 records in 2023 to a peak of 2,551 in 2024, then remained strong in 2025 at 2,177. The 2026 data (41 records) is partial and reflects only the early months of the year. This trend points to consistent and growing international student interest in DePaul graduate programs.

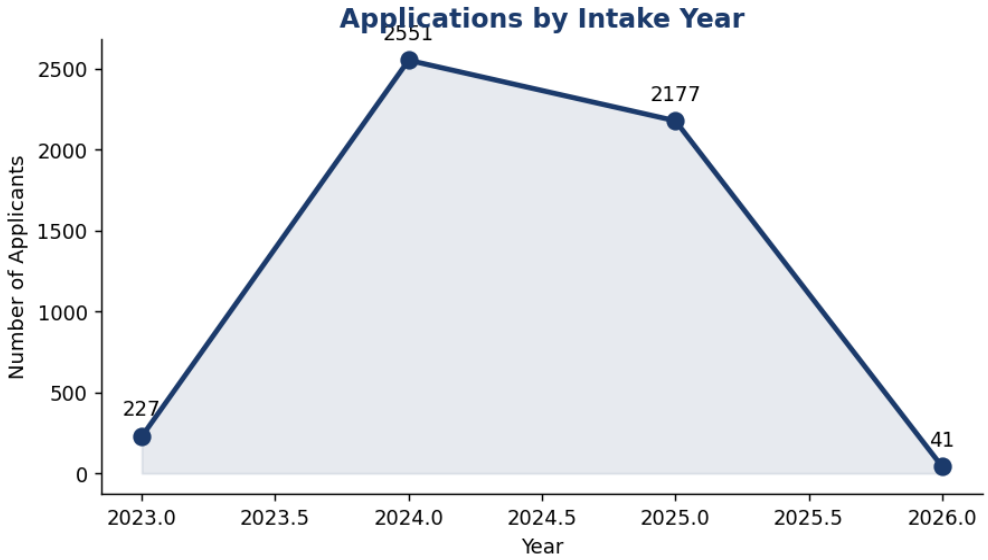



*Figure 6: Application Volume by Intake Year (2023–2026)*

# 5. Insights Summary

The table below consolidates the key findings from the Week 1 data exploration and highlights observations that should guide deeper analysis in Week 2.

| # | Insight | Detail |
|---|---|---|
| 1 | **Top Applicant Country** | India accounts for 53.7% (2,684) of all applicants, far exceeding any other country. The next largest cohort, the United States contributes only 9.8%. |
| 2 | **Dominant College Choice** | Jarvis College of Computing & Digital Media is the most preferred college (48.3% of applicants), reflecting strong global demand for tech-oriented programs. |
| 3 | **Most Popular programs** | Business Analytics and Computer Science together account for 26.3% of all enrolments. Data Science, Health Informatics, and Cybersecurity round out the top five STEM-dominated intakes. |
| 4 | **Seasonal Preference** | Fall semester dominates with 59.7% of all applications (2,983), followed by Winter (25.8%). Summer intake remains minimal at 2.9%. |
| 5 | **Recruitment Channel Split** | 84.2% of applicants came via Non-Study Group channels, while Study Group accounted for 15.8%. This suggests a strong direct or organic pipeline. |
| 6 | **Data Completeness Issue** | A structured block of 3,194 records (63.9%) is missing outreach-related fields (Outcome_1, Caller_Name, Attempts, etc.), indicating these applicants were not contacted or records were not linked. |
| 7 | **Application Volume Trend** | Application volume peaked in 2024 (2,551 records) and remains high in 2025 (2,177), suggesting continued growth in international recruitment. |

## 6. Initial Observations to Inform Week 2 Analysis

- Investigate the 3,194-record outreach gap to determine whether these applicants were intentionally excluded from outreach or represent a data join issue. Cross-referencing with the original source tables will clarify this.
- Perform deeper program-level analysis: explore conversion rates by program, country, and intake year once data is linked across the outreach and application tables.
- Time-series analysis of application volumes for converting epoch timestamps in Received_At and Created_At to readable dates will enable monthly and seasonal trend analysis.
- Outcome analysis: the Outcome_1 field (where populated) captures 17 distinct call outcomes; clustering these into positive, negative and neutral categories could inform outreach effectiveness modelling.
- Country-program affinity mapping: understanding which countries most strongly prefer which programs could inform targeted marketing efforts.

## 7. References & Tools Used

- Dataset: DePaul University Graduate Applicant Dataset (DePaul.xlsx) — provided via Excelerate Internship Program
- Excel — used for data profiling, EDA, and chart generation
- Microsoft Excel — used for initial data inspection and dictionary validation